Assignment No. 15

Compressible flow over an airfoil

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Objective

- Observe solution using pressure based and density based solver for a flow over an airfoil problem

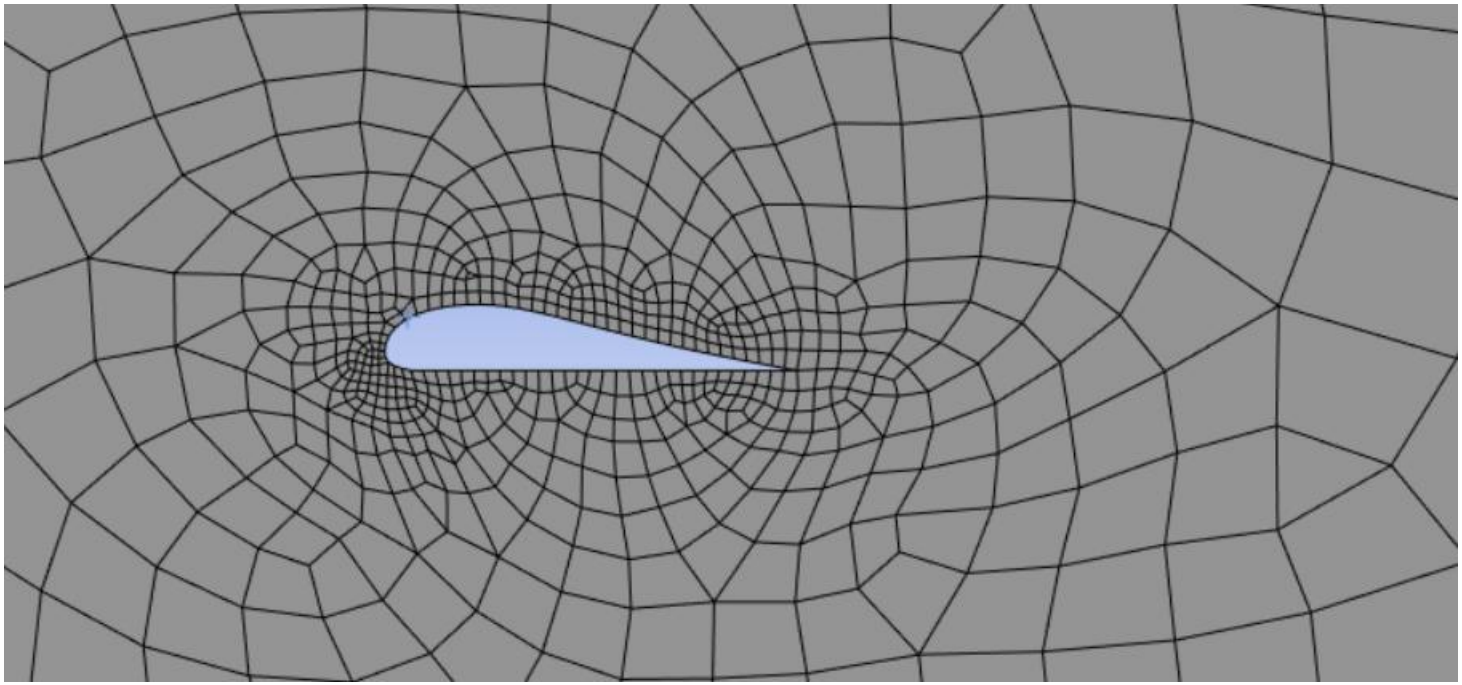
Problem Statement

- Gottigen 1919 airfoil,
- Inlet velocity: 200 m/s, compressible
- Chord length: 32 mm



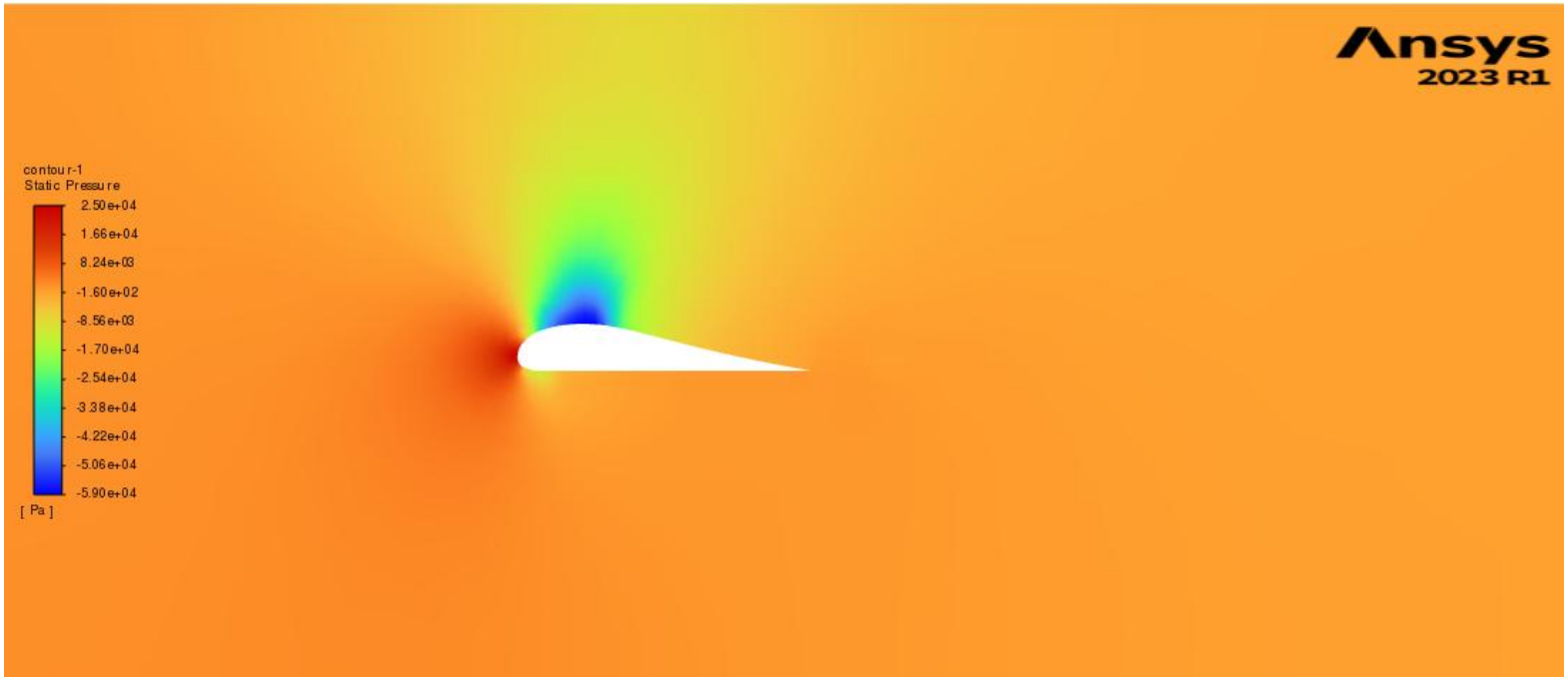
CFD Setting

- Element count: 650
- Material: Air, density: ideal gas
- Inlet velocity: 200 m/s at ambient temperature

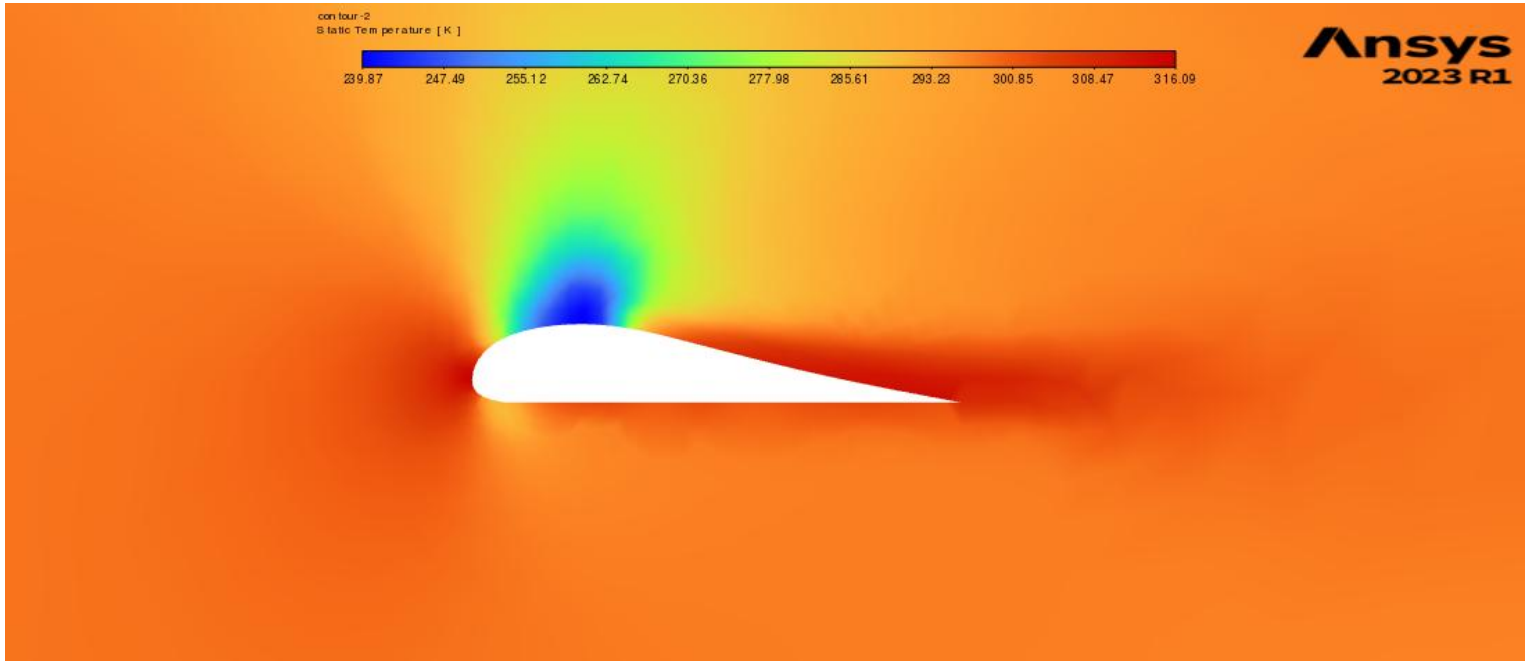


Results(Pressure based-coupled)

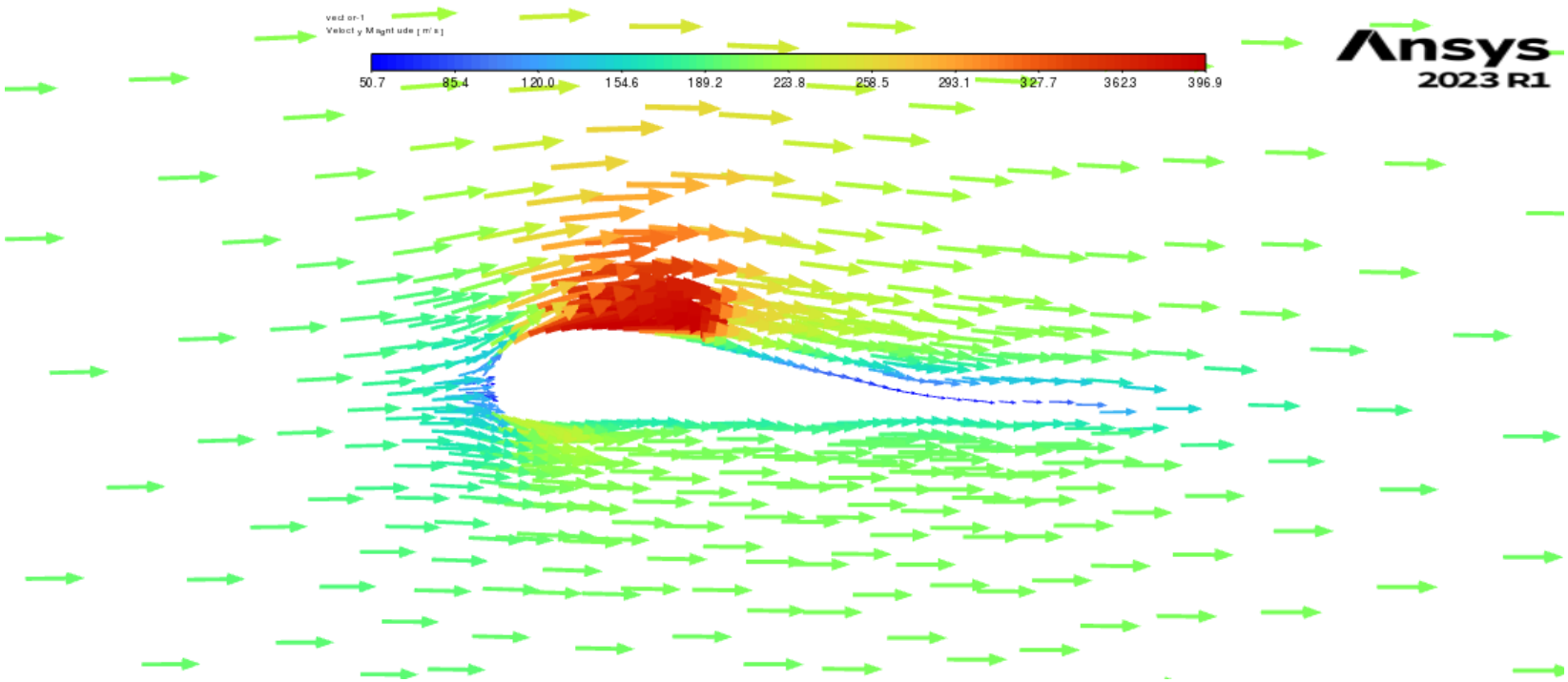
- Solution converged at about 1100 iterations
- Co-efficient of drag: 0.7
- Pressure Contour:



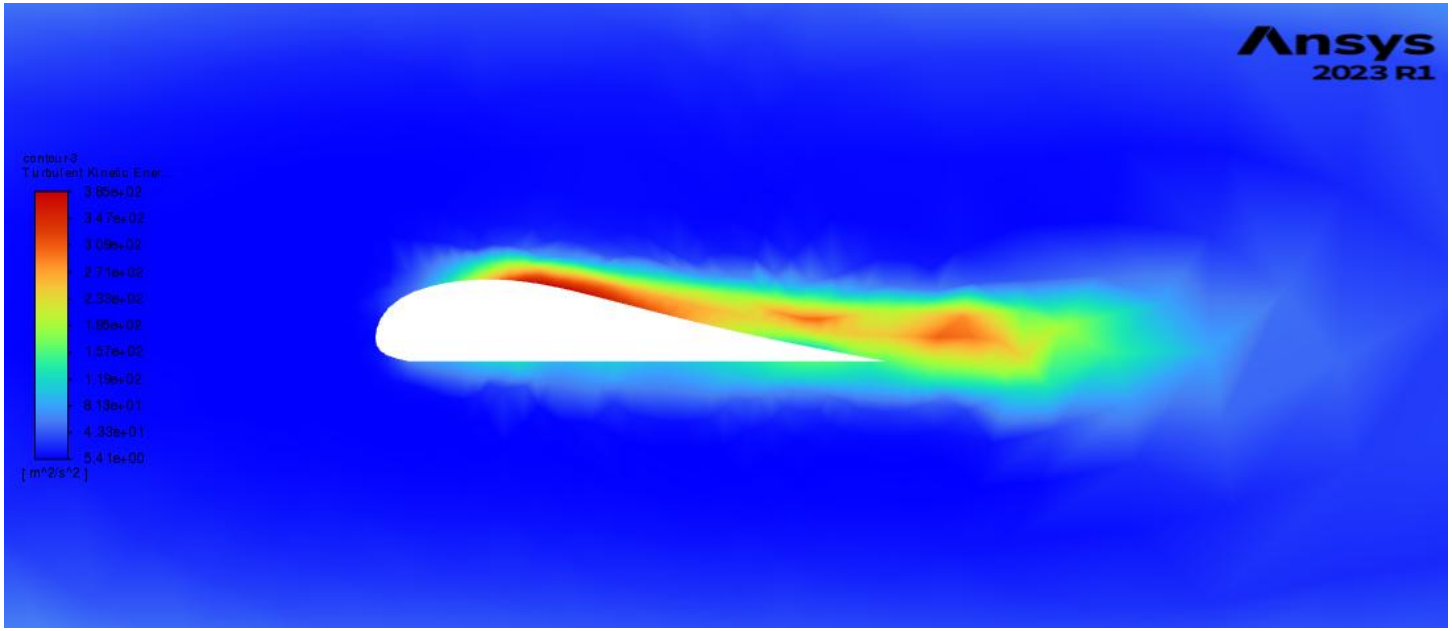
- Temperature contour:



- Velocity Vector:



- TKE contour:


Ansys
2023 R1
contour3
Turbulent Kinetic Ener...
3.85e+02
3.47e+02
3.09e+02
2.71e+02
2.32e+02
1.95e+02
1.57e+02
1.19e+02
8.13e+01
4.33e+01
5.41e+00
[m^2/s^2]

Results(Density based)

- Solution converged at about 8500 iterations
- Co-efficient of drag : 0.8

Conclusion

- Pressure based coupled solver converged earlier with better results